

Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_no_replay.
- Best CVRPLIB holdout gap: cvrp_no_replay.
- Best synthetic holdout gap: cvrp_no_replay.

Run Metadata

- run_name: run_20260513_195210_q
- started_at_local: 2026-05-13 19:52:10
- finished_at_local: 2026-05-13 19:55:13
- duration_hhmm: 00:03
- duration_seconds: 182.675
- seed_offset: 16000
- replicate_label: q
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.203837	0.005376	0.115632	0.862644	0.78	0.017299
cvrp_random_replay	random	score_only	False	0.251603	0.005376	0.142169	0.73025	0.78	0.05445
cvrp_failure_replay	failure	score_only	False	0.215731	0.005376	0.12224	0.82319	0.78	0.0182
cvrp_failure_replay_compression	failure	novelty_gate	True	0.234631	0.005376	0.13274	0.8127	0.78	0.063182

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.203837, synthetic 0.005376, combined 0.115632.

- Accepted-epoch count 2, mean accepted code novelty 0.862644, and final complexity 0.78.
- Adaptation efficiency 0.017299 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.203837 across 5 instances; family means: A=0.081152, B=0.244087, E=0.21881, P=0.231047.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 611, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.334061}, {"best_known_cost": 937, "cost": 1236, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.319104}, {"best_known_cost": 672, "cost": 817, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.215774}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.251603, synthetic 0.005376, combined 0.142169.
- Accepted-epoch count 2, mean accepted code novelty 0.73025, and final complexity 0.78.
- Adaptation efficiency 0.05445 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.251603 across 5 instances; family means: A=0.061955, B=0.24813, E=0.293666, P=0.406137.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 602, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.31441}, {"best_known_cost": 937, "cost": 1219, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.300961}, {"best_known_cost": 672, "cost": 805, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.197917}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.215731, synthetic 0.005376, combined 0.12224.
- Accepted-epoch count 2, mean accepted code novelty 0.82319, and final complexity 0.78.
- Adaptation efficiency 0.0182 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.215731 across 5 instances; family means: A=0.081152, B=0.244086, E=0.247601, P=0.261733.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 611, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.334061}, {"best_known_cost": 937, "cost": 1236, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.319104}, {"best_known_cost": 672, "cost": 805, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.197917}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: CVRPLIB 0.234631, synthetic 0.005376, combined 0.13274.

- Accepted-epoch count 2, mean accepted code novelty 0.8127, and final complexity 0.78.
- Adaptation efficiency 0.063182 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.234631 across 5 instances; family means: A=0.145724, B=0.21309, E=0.214971, P=0.386282.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1219, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.300961}, {"best_known_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.200893}]

Judge Appendix

Conservative Interpretation of Replay-Aware CVRP Benchmark Results

Metric	No Replay (cvrp_no_replay)	Random Replay (cvrp_random_replay)	Failure Replay (cvrp_failure_replay)	Failure Replay + Compression (cvrp_failure_replay_compression)
Held-out CVRPLIB mean gap	**0.2038**	0.2516	0.2157	0.2346
Synthetic holdout mean gap	0.0054	0.0054	0.0054	0.0054
Mean transfer gap	**0.1156**	0.1422	0.1222	0.1327
Code novelty (mean)	0.8626	0.7303	0.8232	0.8127
Compression pressure	No	No	No	Yes
Replay mode	None	Random	Failure	Failure + Compression
Selection mode	score_only	score_only	score_only	novelty_gate

Key Observations

- ****Transfer Performance****
- ***Best held-out CVRPLIB gap (0.2038)*** and ***best mean transfer gap (0.1156)*** are achieved by the ****no replay**** condition.
- All replay modes (random, failure, failure + compression) show ****worse transfer gaps**** and held-out CVRPLIB gaps relative to no replay.
- Synthetic holdout gaps are uniformly low (0.0054) across all conditions, indicating minimal differences on synthetic benchmarks.
- This supports concluding that ****no replay yields better transfer performance**** on held-out CVRPLIB instances and synthetic holdouts.

- **Code Novelty vs. Transfer**
- No replay achieves **highest lexical code novelty (0.8626)** and also the best transfer.
- Replay conditions reduce code novelty (0.73 to 0.82) but do **not improve transfer**; in fact, transfer worsens despite some adaptation efficiency gains under replay.
- Hence, **code novelty decline with replay modes is associated with transfer degradation**, refuting the idea that replay improves algorithmic generalization.
- **Effect of Compression-aware Replay**
- Compression pressure is applied only in failure replay with compression.
- This condition improves adaptation efficiency (0.0632; highest) but yields worse transfer gaps than no replay or failure replay without compression.
- This indicates that **compression-aware replay may aid adaptation speed but at a cost to transfer optimality**.
- **Optimality Gap Breakdown (Held-out CVRPLIB)**
- Worst gaps appear in families B, P, E, with family P gaps particularly large under replay conditions (up to 0.4061).
- No replay shows more balanced family gaps and lower worst-case gaps.
- **Behavior and Complexity**
- Across conditions, final behavior profiles (clustered_constructor) and complexity (~0.78) remain consistent, suggesting stable algorithmic structure regardless of replay.

Summary

- The **no replay condition achieves best transfer results**, demonstrating superior generalization from training to held-out CVRPLIB and synthetic instances.
- Introduction of **random, failure, or compression-aware replay reduces transfer quality**, despite some gains in adaptation efficiency.
- **Code novelty decreases with replay use; this decrease parallels worse transfer**, suggesting replay harms algorithmic innovation relevant to generalization.
- Improvements in adaptation efficiency under replay scenarios do not translate into improved optimality gaps on held-out benchmarks.
- Narrative claims that replay enhances transfer or algorithmic discovery are **not supported by the deterministic optimality gap and transfer metrics**.

Recommendations

- Prioritize the **no replay approach** when aiming for optimality and transfer generalization.
- Use caution when introducing replay mechanisms; they may incur transfer cost despite adaptation speed benefits.
- Future work should investigate why replay diminishes transfer and whether alternative replay strategies (beyond current modes) might preserve code novelty and improve transfer gaps.